

1 Voting Theory

1.4 Borda Count Method and Voting Theory Review

Example 1.4.1 Let’s just dive right into an example.

	51	25	10	14
1st	S	T	P	O
2nd	T	P	T	T
3rd	O	O	O	P
4th	P	S	S	S

Go over how points are assigned, then do the example. T is the winner of this election, with 325 points.

Example 1.4.2 Determine who wins under the Borda Count method.

	44	14	70	22	80
1st	G	G	M	M	B
2nd	M	B	G	B	M
3rd	B	M	B	G	G

Example 1.4.3 Your turn! Use the Borda Count method to determine who wins the following two elections!

	11	5	10	3
1st	A	C	B	A
2nd	C	B	C	B
3rd	B	A	A	C

Example 1.4.4 Use the Borda Count method to determine who wins the following election!

	120	50	40	90
1st	C	B	D	A
2nd	D	C	A	C
3rd	B	A	B	B
4th	A	D	C	D

Example 1.4.5 Let's go back to the first example. Recall that the winner of the election was T with 325 points. Is there something not fair about that?

	51	25	10	14
1st	S	T	P	O
2nd	T	P	T	T
3rd	O	O	O	P
4th	P	S	S	S

S has a majority!

Example 1.4.6 Use the Plurality method to determine the winner of this election.

	9	19	11	8
1st	B	A	C	B
2nd	A	B	B	C
3rd	C	C	A	A

Example 1.4.7 With the same preference schedule as in the previous example, who wins under the Pairwise Comparison method?

Example 1.4.8 Use the Plurality with Elimination method to determine who wins this election.

	120	50	40	90	60	100
1st	C	B	D	A	A	D
2nd	D	C	A	C	D	B
3rd	B	A	B	B	C	A
4th	A	D	C	D	B	C

Example 1.4.9 Time to think! Does the Pairwise Comparison Method Satisfy the Majority Criterion? Why or why not?

Fairness Criterion: These are all things that we think should be true if a voting method is fair.

1. Condorcet Criterion: If there is a Condorcet candidate, they should win the election.
2. Independence of Irrelevant Alternatives (IIA) Criterion: if a loser drops out of the election, this should not hurt the winners chances of winning.
3. Monotonicity Criterion: If voters change their votes to support the winner, this should not hurt the winners chances of winning.
4. Majority Criterion: If a candidate has a majority, they should win the election.

Theorem 1.4.10 (*Arrow's Impossibility Theorem*) *No voting method is guaranteed to always satisfy all four fairness criterion.*